

The Minimal Rating Process: Sovereign Credit Ratings from Risk-Off Positioning

MRP specification, v1.0 — for discussion

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August 2026

Abstract

This document specifies the *minimal rating process* (MRP): a complete, streamlined procedure for rating the credit risk of euro-area sovereigns, built on the model of Stomper (2026, “the Note,” v7). The organizing idea: countries differ in how they are positioned for *risk-off* — episodes in which the market’s willingness to bear risk falls — and this positioning determines sovereign credit risk. The process consists of six **calibration tasks**, which set the model’s parameters from public data once a year, and seven **rating tasks**, which implement the model at monthly frequency and generate the rating; the rating tasks are the core of the process. Every computation is a closed form or a one-dimensional solve; every input is a published number or a sample moment; the whole process fits on one page and reproduces exactly from the published parameter sheet. The MRP is specified for the case without captive creditors — the conservative benchmark, since captivity narrows the fragile region — and treats inflation through the one variable the model says matters: the expected inflation differential between the member state and the union.

1 Purpose, philosophy, scope

A sovereign rating is an ordinal opinion about the probability of default over a horizon. The MRP forms that opinion from a country’s *position relative to a fold in its market access*: how large a risk-off move — formally, an increase in the market price of distress risk θ defined in Section 2 — would leave the country with only a crisis equilibrium, and how much fiscal effort would take it out of danger under its own control. Four principles, carried over from the underlying theory, discipline the process:

1. **Rate the coordinates, not the price.** Inside the model’s fragile region two market spreads are consistent with the same fundamentals, so the spread cannot be inverted into risk there. The MRP rates the country’s coordinate (n_t, θ_t) relative to the fold — and may therefore legitimately disagree with market prices, in both directions, in a way the process makes precise.
2. **Measure positioning in units of one global factor.** All distances are denominated in units of θ , the same axis for every country, and convert into probabilities under the factor’s distribution. Comparability across countries comes from a common state variable, not a common checklist.
3. **Replicability.** The rating is a deterministic function of a published parameter vector; given the parameter sheet and the public code, any reader reproduces it exactly.
4. **Discretion calibrates or warns; it never scores.** Judgment may inform the calibration tasks (within documented ranges, with published justification) and the outlook; it may not move a class. The MRP has no notches.

Scope. Euro-area member states: sovereigns that borrow in a unit of account they cannot print, so that a refusal of the market to lend can force default and the fragile-region apparatus applies with full force. Extensions (own-currency sovereigns, emerging markets) are outside this document.

No captive creditors. The Note’s v7 models a *captive creditor* layer — creditors who roll over a fixed stock at a regulated price irrespective of default risk. The MRP sets this layer to zero ($k = 0$). This is a deliberate, conservative choice, not a simplification of convenience: the Note’s Proposition 2 shows that a captive tranche raises both edges of the fragile band and *narrows* it — beliefs can only reprice the market-held tranche — so the $k = 0$ benchmark is the fragility-maximal case. Rating at $k = 0$ errs on the side of caution, and it removes the need to measure the captive stock and its regulatory wedge. Where a country’s captivity is material, the $k = 0$ rating is a floor on its positioning, and the fact is noted in the outlook, not in the class.

2 The model in twelve equations

Notation: all rates and growth factors are *gross* ($R^f = 1.02$ means 2 percent); Φ and ϕ are the cdf and density of the standard normal distribution — the only distribution used; t indexes years.

Growth and inflation. Nominal GDP grows at the gross rate G_{t+1} , i.i.d. lognormal,

$$\ln G_{t+1} \sim N(\mu, \sigma^2), \quad F(x) = \mathbb{P}[G \leq x] = \Phi\left(\frac{\ln x - \mu}{\sigma}\right). \quad (1)$$

Nominal growth bundles real growth with domestic inflation, and the safe rate bundles the real rate with union-wide inflation:

$$\mu = \mu_r + \pi^d, \quad \ln R^f = r + \pi^u, \quad e \equiv \pi^d - \pi^u, \quad (2)$$

where μ_r is the real log-growth drift, π^d expected domestic inflation, π^u expected union-wide inflation (to which the union safe asset is Fisher-linked), and e the expected **inflation differential**. The Note’s Proposition 3 establishes that the model is *exactly neutral* with respect to the common component of anticipated inflation: shifting π^d and π^u together leaves every real magnitude — capacity, band edges, spreads, distances, exposure — unchanged. At $k = 0$ this neutrality is unconditional. Everything inflation-related therefore runs through the single variable e ; Section 3 (Task C2) states how the MRP measures it and why conservatively.

The refinancing need. The government can sustain a primary surplus of at most $\bar{s}$ per year (share of GDP) and follows a plan $\{s_t\}$, $s_t \leq \bar{s}$. Debt is a one-period zero-coupon bond, d_t the face value due at $t+1$ per unit of date- t GDP; in default, face values are written down by the haircut $h \in (0, 1)$. At date t the government must raise the **refinancing need**

$$n_t = \frac{d_{t-1}}{G_t} - s_t, \quad (3)$$

an accounting identity; with a maturity structure this is the published *gross financing need* (GFN). The model’s short-run state variable is observable.

Pricing and the funding curve. A competitive lender prices a unit of face value that defaults with probability p at $q(p) = (1 - hp)/R^f$; the promised return and spread are

$$R = \frac{R^f}{1 - hp}, \quad \psi = R - R^f = \frac{R^f hp}{1 - hp} \approx hp. \quad (4)$$

With the default probability depending on the face value issued through a nondecreasing function P ($P(0) = 0$), auction proceeds define the **funding curve**

$$\xi(d) = \frac{d(1 - hP(d))}{R^f}, \quad (5)$$

and an equilibrium of the date- t auction solves $\xi(d) = n_t$. Proceeds are non-monotone — ξ has a local maximum followed by a local minimum — if and only if

$$h[P(d) + dP'(d)] > 1 \quad \text{for some } d \quad (\text{the dip condition}). \quad (6)$$

Under (6), let $d_U < d_L$ be the interior stationary points of ξ and

$$n_U = \xi(d_U), \quad n_L = \xi(d_L) \quad (7)$$

the proceeds at the local maximum and minimum. Three regimes: **safe** ($n_t < n_L$: unique equilibrium, near-zero spread); **fragile** ($n_L \leq n_t \leq n_U$: a good equilibrium G , an unstable one, and a bad equilibrium B with spreads up to the recovery ceiling $R^f h/(1 - h)$ coexist, selected by beliefs); **distressed** ($n_t > n_U$: only B).

Capacity. Default at $t+1$ occurs iff growth is too low for the need to be coverable given **maximum sustainable borrowing** b_M — the largest amount the market lends against future surpluses and rollovers (Collard–Habib–Rochet). With γ the default-risk-discounted certainty-equivalent growth factor,

$$P(d) = F\left(\frac{d}{\bar{s} + b_M}\right), \quad \gamma = \max_{\chi \geq 0} (1 - F(\chi))\chi, \quad b_M = \frac{\bar{s}\gamma}{R^f - \gamma}, \quad (8)$$

and $d_M = (\bar{s} + b_M)\chi_M$ (the maximizer χ_M) is the maximum face value — “the cliff.” Because $\gamma < R^f$ at every relevant calibration, capacity is finite even when $R^f < \mathbb{E}[G]$. Under lognormality, with z_M the unique negative solution of $\sigma(1 - \Phi(z_M)) = \phi(z_M)$,

$$\gamma(\theta) = (1 - \Phi(z_M))e^{\mu - \theta\sigma + \sigma z_M}, \quad b_M(\theta) = \frac{\bar{s}\gamma(\theta)}{R^f - \gamma(\theta)}, \quad d_M(\theta) = (\bar{s} + b_M(\theta))e^{\mu - \theta\sigma + \sigma z_M}, \quad (9)$$

where θ is the market price of distress risk introduced next, and the default-probability function is

$$P_\theta(d) = \Phi\left(\frac{\ln d - \ln(\bar{s} + b_M(\theta)) - (\mu - \theta\sigma)}{\sigma}\right). \quad (10)$$

Risk appetite. Sovereign default happens in global bad states, so default risk carries a premium: a one-factor pricing kernel loading on standardized log growth shifts the pricing drift from μ to $\mu - \theta_t\sigma$, where $\theta_t \geq 0$ is the **market price of distress risk** — the model’s representation of global risk appetite (θ high = risk-off). θ_t is a price, not directly observable; rating task R2 estimates it monthly, so the country’s coordinate $(n_t, \hat{\theta}_t)$ is refreshed every month. It reverts to a long-run anchor θ_∞ ; the MRP treats risk-off moves as permanent ($\rho_\theta = 1$ in the Note’s notation), the conservative choice, so (9)–(10) are evaluated at $\theta = \hat{\theta}_t$ throughout. Writing ξ_θ , $n_L(\theta)$, $n_U(\theta)$ for the funding curve and edges at these risk-adjusted objects, both edges fall as θ rises: *risk-off shifts the whole funding curve down*. For a country with need n , the **critical risk prices**, **exit distances**, and **exposure probability** are

$$\theta_G(n) : n_U(\theta_G) = n, \quad \theta_B(n) : n_L(\theta_B) = n, \quad (11)$$

$$\Delta_G = \theta_G(n_t) - \theta_t, \quad \Delta_B = \theta_t - \theta_B(n_t), \quad (12)$$

$$\text{Exposure}_{it} = \mathbb{P}[\theta > \theta_G(n_{it})] = 1 - \Phi\left(\frac{\theta_G(n_{it}) - \theta_\infty}{\sigma_\theta}\right) \quad (13)$$

under the factor distribution $N(\theta_\infty, \sigma_\theta^2)$ delivered by task R2. The good equilibrium exists iff $\theta_t \leq \theta_G(n)$, the bad iff $\theta_t \geq \theta_B(n)$; Exposure is the probability of a risk-appetite realization that leaves the country with only the bad equilibrium — a well-defined tail probability, not a heuristic index. Two properties of the theory shape the rating tasks: *persistence is key* (a risk-off episode is dangerous in proportion to how permanent the market believes it to be — the reason the MRP’s $\rho_\theta = 1$ is the conservative choice), and *hysteresis* (once in B , returning to G requires the much stronger $\theta_t < \theta_B(n_t)$, or a credible *backstop* — a commitment by an institution whose resources the market cannot exhaust, in practice the ECB or ESM, to buy or guarantee the debt under stated conditions, which deletes B from the equilibrium set without any purchases).

3 Calibration tasks (annual)

Six tasks set the model’s parameters $(\mu, \sigma, \bar{s}, R^f, h; \theta_\infty, \sigma_\theta)$ from public data, re-run annually or on major fiscal events. This is where soft information may enter — always within documented ranges, with published justification (Principle 4). All sources are free.

Task C1 (real growth moments).

Sample moments of log *real* GDP growth g_t^r over the last $T = 25$ years (IMF WEO):

$$\hat{\mu}_r = \frac{1}{T} \sum_{t=1}^T g_t^r, \quad \hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (g_t^r + \hat{\pi}_t^d - \hat{\mu}_r - \hat{\pi}^d)^2, \quad (14)$$

where the volatility is computed on nominal growth (real growth plus realized inflation $\hat{\pi}_t^d$), because default is triggered by nominal shortfalls. No filtering: the COVID and inflation years are information about σ , not outliers.

Task C2 (inflation: the differential e).

The model needs an inflation stance only through $e = \pi^d - \pi^u$ (2): the common component cancels exactly (Proposition 3 of the Note), so the MRP takes no view on union-wide inflation — it is inherited automatically from the market R^f . What remains is measured and used as follows. Let $\hat{e}_t$ be the projected five-year average HICP inflation differential between the member state and the euro area (WEO or ECB projections). The MRP uses

$$e_{\text{MRP}} = \min\{0, \hat{e}_t\}, \quad \hat{\mu} = \hat{\mu}_r + \hat{\pi}^u + e_{\text{MRP}}, \quad (15)$$

with $\hat{\pi}^u$ the projected five-year euro-area HICP average. The asymmetry is the task’s substance. A *positive* differential (domestic inflation above the union’s) raises capacity in the model — it erodes debt faster than the union-linked discount rate compensates — but the Note’s own caveat applies: in a currency union, inflation differentials mean-revert through competitiveness, so crediting a current positive differential as permanent overstates capacity; the MRP credits none of it. A *negative* differential (expected lowflation or deflation relative to the union) destroys capacity at a first-order rate of

$$\frac{\partial b_M}{\partial e} = \frac{\gamma \bar{s} R^f}{(R^f - \gamma)^2} \quad (\approx 6.3\text{pp of GDP per 100bp at the reference calibration of Section 6), \quad (16)$$

and this is the euro-area inflation configuration that actually threatens credit risk; the MRP debits it in full. Sensitivity in $e \in \{\hat{e}_t, 0\}$ is always published.

Task C3 (fiscal capacity $\bar{s}$ and plan s_t).

The best sustained fiscal performance in the country's own history (WEO/Fiscal Monitor primary balances pb_j , share of GDP),

$$\bar{s} = \max_t \frac{1}{5} \sum_{j=t-4}^t pb_j, \quad (17)$$

one number, defensible in one sentence: the country has demonstrated it can sustain this. The plan s_t is the current-year projected primary balance. Soft information (institutional quality, surplus fatigue, restructuring history) may place $\bar{s}$ below (17) with published justification; it may never raise it above the demonstrated maximum.

Task C4 (safe rate R^f).

The one-year point of the ECB's euro-area AAA yield curve, monthly average, expressed gross.

Task C5 (haircut h).

Fixed at 0.30, the center of the historical distribution of restructuring haircuts (Cruces–Trebesch database); the sensitivity grid of task R7 always spans $h \in \{0.20, 0.50\}$.

Task C6 (risk-appetite parameters).

The anchor and dispersion of the distress-risk price, $(\theta_\infty, \sigma_\theta) = (0.30, 0.25)$: the anchor is the Note's conservative long-run level, consistent with the Longstaff–Pan–Pedersen–Singleton finding that roughly one-third of the level of sovereign spreads is a distress-risk premium; the dispersion scales one standard deviation of the observable proxy (task R2) into kernel units, consistent with the Note's calibrated illustrations. Alongside, compute the 20-year mean and standard deviation of the log monthly VIX, (m_V, s_V) , used by task R2. These four numbers are calibration constants: documented, contestable, and revised only with a published note.

4 Rating tasks (monthly) — the core of the process

Seven tasks implement the model at the calibrated parameters and generate the rating. They run every month; each monthly run is a potential rating action.

Task R1 (measure the refinancing need n_t).

The gross financing need from four published numbers — debt ratio d_{t-1}^{stock} , short-term (under-one-year) debt stock b_{t-1}^{ST} , weighted average maturity M_{t-1} of the remaining portfolio, headline deficit def_t (all shares of GDP):

$$n_t \approx \frac{d_{t-1}^{stock} - b_{t-1}^{ST}}{M_{t-1}} + b_{t-1}^{ST} + def_t, \quad (18)$$

cross-checked against the GFN published in the country's IMF Article IV tables or the Commission's Debt Sustainability Monitor. Sources for the components: WEO/Eurostat and the national debt-management office.

Task R2 (estimate the market price of distress risk $\hat{\theta}_t$).

The object is the kernel coefficient of Section 2: the wedge, in growth-standard-deviation units, between the physical drift and the drift at which the marginal investor prices default risk — global, common to all rated countries. The MRP estimator is a two-step plug-in. First, standardize the log of the monthly average VIX V_t against its own 20-year history (constants m_V, s_V from C6):

$$z_t = \frac{\ln V_t - m_V}{s_V}; \quad (19)$$

second, map affinely into kernel units:

$$\hat{\theta}_t = \max \{0, \theta_\infty + \sigma_\theta z_t\}. \quad (20)$$

Because z_t is approximately standard normal by construction, (20) also delivers the factor distribution $N(\theta_\infty, \sigma_\theta^2)$ used in (13). The estimator's simplicity is deliberate: one observable, one monotone map, both constants calibrated in C6 and published.

Task R3 (capacity objects).

Solve $\sigma(1 - \Phi(z_M)) = \phi(z_M)$ for z_M ; compute $\gamma(\hat{\theta}_t)$, $b_M(\hat{\theta}_t)$, $d_M(\hat{\theta}_t)$ by (9) and the function P_θ by (10), at the calibrated $(\hat{\mu}, \hat{\sigma}, \bar{s}, R^f)$.

Task R4 (band edges and regime).

Solve $\xi'_\theta(d) = 0$ for the stationary points of the funding curve (5) evaluated with P_θ , and obtain $n_L(\hat{\theta}_t)$, $n_U(\hat{\theta}_t)$ by (7). Classify: **safe** ($n_t < n_L$), **fragile** ($n_L \leq n_t \leq n_U$), **distressed** ($n_t > n_U$). Inside the fragile region, identify the prevailing equilibrium from the observed spread: good-branch pricing (a few bp to ~ 100 bp) versus bad-branch pricing (approaching $R^f h / (1 - h)$, by (4) at $p \rightarrow 1$); the branches are far apart, so the identification is robust. As a diagnostic, compare the observed 10-year spread with the model's default component $\psi \approx h P_\theta(d_t)$ at $d_t \approx n_t R^f$: for a safe country the model component is ≈ 0 and the observed spread is liquidity, not default risk.

Task R5 (exits).

Solve (11) for $\theta_G(n_t)$, $\theta_B(n_t)$; compute Δ_G , Δ_B by (12), Exposure by (13), and the **fiscal exit time**

$$X_{it}^{fisc} = \frac{\max\{n_{it} - n_L(\theta_\infty), 0\}}{\bar{s} - s_{it}} \quad (21)$$

— the years of maximum additional fiscal effort needed to leave the band downward: the own-control exit in effort units, complementing the market-controlled exit priced by Δ_G in factor units. For distressed countries, compute additionally the risk-on exit probability $\mathbb{P}[\theta < \theta_B(n_t)]$.

Task R6 (rating class).

The class follows from thresholds with probability semantics; there are no notches.

safe:	S1: Exposure $\leq 0.01\%$	S2: Exposure $\leq 0.1\%$	S3: Exposure $\leq 1\%$	
fragile (G):	F1: Exposure $\leq 5\% \wedge X_{it}^{fisc} \leq 2y$	F2: Exposure $\leq 15\%$	F3: Exposure $\leq 30\%$;	F4: $> 30\%$
distressed/ B :	D1: $\mathbb{P}[\theta < \theta_B] \geq 25\%$	D2: $< 25\%$	D3: $n_t > n_U(\theta)$ on the 90% ra	

(Letter territory, for communication only: S1 $\approx$ AAA/AA+ down to D3 $\approx$ imminent SD.) Two rules complete the task. *Backstop eligibility*: determine the state $e_{it} \in \{\text{eligible, watch, ineligible}\}$ from observable bright lines (clean compliance with the EU fiscal framework and a low-risk Commission DSA classification $\Rightarrow$ eligible; an open excessive-deficit procedure with effective action established $\Rightarrow$ watch; non-compliance $\Rightarrow$ ineligible). If *eligible*, the Exposure input above is computed with the factor distribution truncated at $\theta_G(n_t)$: a credible conditional backstop deletes the bad equilibrium for rating purposes, moving the class through the measured score, never past it. *Hysteresis*: downgrades into D follow the classification instantly; upgrades out of D require $\hat{\theta}_t < \theta_B(n_t)$ for three consecutive months, or a backstop/program event — admissible only in state *eligible* — that deletes B .

Task R7 (outlook and publication).

The **outlook** $\in \{\text{positive, stable, negative}\}$ is the one discretionary output: its base input is the sign of the projected twelve-month change in Δ_G (or in $\mathbb{P}[\theta < \theta_B]$ for D countries) along the announced fiscal path at constant θ , applying (18) to budget plans

and the redemption calendar; information beyond the model (backstop politics, banking stress, investor-base shifts, a material captive layer) may adjust the outlook — never the class — with published justification. The **publication sheet** contains: the calibrated parameters ($\hat{\mu}_r, \hat{\pi}^u, e_{\text{MRP}}, \hat{\sigma}, \bar{s}, s_t, R^f, h, \theta_\infty, \sigma_\theta$); the state ($n_t, \hat{\theta}_t$); the derived objects ($z_M, \gamma, b_M, d_M, n_L, n_U, \theta_G, \theta_B, \Delta_G, \Delta_B, \text{Exposure}, X^{\text{fisc}}$); regime, class, eligibility state, outlook with rationale; a **sensitivity grid** over $\bar{s} \times h \times \hat{\theta}_t \times n_t \times e$ (three values each where applicable), reporting the regime count and the worst-corner Exposure — the MRP’s honest substitute for formal uncertainty quantification; and the code version.

5 The process on one page

Task	Output	Eq.	Input (source)
<i>Calibration (annual)</i>			
C1	$\hat{\mu}_r, \hat{\sigma}$	(14)	real/nominal GDP growth, 25y (IMF WEO)
C2	$e_{\text{MRP}}, \hat{\mu}$	(15)	HICP projections, state vs. euro area (WEO/ECB)
C3	$\bar{s}, s_t$	(17)	primary balances (WEO/Fiscal Monitor)
C4	R^f	—	1y euro-area AAA yield (ECB)
C5	$h = 0.30$	—	Cruces–Trebesch haircut distribution
C6	$(\theta_\infty, \sigma_\theta); (m_V, s_V)$	—	Note calibration; 20y log-VIX history (CBOE)
<i>Rating (monthly) — the core</i>			
R1	n_t	(18)	debt ratio, bills, WAM, deficit (WEO/DMO); GFN cross-check
R2	$\hat{\theta}_t; \theta \sim N(\theta_\infty, \sigma_\theta^2)$	(19)–(20)	monthly VIX (CBOE)
R3	$z_M, \gamma, b_M, d_M, P_\theta$	(9)–(10)	C1–C5 outputs, $\hat{\theta}_t$
R4	n_L, n_U ; regime, branch	(7)	funding curve (5); 10y spread (ECB/DMO)
R5	$\theta_G, \theta_B, \Delta_G, \Delta_B, \text{Exposure}, X^{\text{fisc}}$	(11)–(13), (21)	R1–R4 outputs
R6	class; eligibility state	thresholds	Exposure, X^{fisc} ; EDP/DSA status (Commission)
R7	outlook; publication sheet	—	fiscal plans, redemption calendar; grid

Table 1: The minimal rating process: six calibration tasks, seven rating tasks.

6 Worked illustration

At the Note’s reference calibration for a typical euro-area member ($\mu = 3\%$, $\sigma = 3\%$, $\bar{s} = 4\%$, $R^f = 1.03$, $h = 30\%$, plan $s_t = 2\%$; $k = 0$): R3 gives $z_M = -2.28$, $\gamma = 0.951$, $b_M = 48.5\%$, $d_M = 50.5\%$ of GDP; R4 gives, at $\hat{\theta}_t = 0$, $n_L = 39.4\%$ and $n_U = 49.2\%$. A country with $n_t = 37\%$ is safe at $\hat{\theta}_t = 0$ but fragile for $\theta \in [0.16, 0.83]$: R5 gives $\theta_B = 0.16$, $\theta_G = 0.83$. If R2 delivers $\hat{\theta}_t = 0.30$, the country sits inside the band on good-equilibrium pricing, with $\Delta_G = 0.53$, $\text{Exposure} = 1 - \Phi((0.83 - 0.30)/0.25) \approx 1.7\%$, and $X^{\text{fisc}} = (0.37 - 0.352)/(0.04 - 0.02) \approx 0.9$ years: class **F1** — fragile, but with a modest tail probability and an exit within its own fiscal control. The same country at $n_t = 45\%$ would show $\text{Exposure} > 30\%$ and no quick fiscal exit: class **F4** — *at an unchanged debt ratio*. The debt level does not determine the rating; the position relative to the fold does. On the inflation side, (16) evaluates to 6.3: a member state expected to run inflation 100bp *below* the union average for the projection horizon loses about 6 points of GDP of borrowing capacity — roughly, both band edges shift down by a comparable

magnitude — which is why Task C2 debits negative differentials in full while crediting positive ones not at all.

7 What this document deliberately omits

The MRP is the skeleton. Refinements exist for every task — estimated factor dynamics and persistence in place of the calibrated map (19)–(20), Bohn-regression and realism cross-checks for $\bar{s}$, the Commission-DSA maturity machinery in place of (18), parameter bootstraps in place of the sensitivity grid, country-specific factor loadings from investor-base data — and are documented separately, each defined by the basic-task output it modifies. The captive-creditor layer of the Note (which narrows the fragile band and adds an extraction flow to capacity) is switched off here by the conservative $k = 0$ choice; a rating variant that credits measured captivity is likewise a documented extension, not part of the MRP. The relation of the approach to existing standards (IMF SRDSF, the Commission DSA and Debt Sustainability Monitor, the OECD Global Debt Report, and the academic benchmarks) is maintained in a separate companion document.

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